

MUGE Final 7-System Canonical: 10-Term Resonance Acceleration Suite

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Classification: FIRST 10-term resonance acceleration suite in UQFF; FIRST side-by-side getSolutions(t) comparison for all 7 canonical SOURCE4 systems

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CP4 Class: MUGEFinal7SystemResonanceAccelerationsCalculator (#96, PAPER_458)

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $[SCm] = 0.99$ -->

Abstract

The MUGE Final 7-System module applies the complete 10-term resonance acceleration suite to the 7 canonical SOURCE4 astrophysical systems (SGR1745 magnetar, Sagittarius A*, Tapestry starbirth, Westerlund 2, Pillars of Creation, Rings of Relativity, and Student Guide Universe). Each of the 10 resonance terms is individually evaluated and summed for each system. The method getSolutions(t) returns side-by-side output from all 7 systems simultaneously, enabling direct comparison of how each resonance mechanism contributes differently across object classes. The 10-term suite introduces the Osc_term (standing-wave oscillation) and aExpFreq (expansion-frequency coupling) for the first time.

2. The 10-Term Resonance Acceleration Suite — PAPER_458

2.1 Term Listing

#	Term	Symbol	Formula
1	THz hole coupling	a_{THz}	$c^3/(GMr) \times f_{\text{THz}}^2$
2	Vacuum differential	$a_{\text{vac_diff}}$	$\rho_{\text{vac}}[SCm] \times V^{(1/3)} - \rho_{\text{vac}}[UA] \times V^{(1/3)}$
3	Super-frequency	$a_{\text{SuperFreq}}$	$\sum A_k \sin(2\pi f_k t)$, $k=1..5$
4	Aether resonance	$a_{\text{AetherRes}}$	$\rho_{\text{vac}}[SCm](1+[SSq]^{(n26-1)}) V_{\text{sys}}^{(1/3)}$
5	Ug4 vacuum	$Ug4_i$	$U_A \rho_{\text{vac}}(1+[UA][SCm])$
6	Quantum frequency	$a_{\text{QuantumFreq}}$	$\omega_q / (M c^2 r) \times c$
7	Aether frequency	$a_{\text{AetherFreq}}$	$f_{\text{aether}} \times r \times [SCm]$
8	Fluid frequency	$a_{\text{FluidFreq}}$	$\nu_{\text{fluid}} \times f_{\text{fluid2}} \times r$
9	Oscillation standing wave	Osc_term	$A_{\text{osc}} \cos(k_{\text{osc}} r) \sin(\omega_{\text{osc}} t)$
10	Expansion frequency	a_{ExpFreq}	$H_0 \times c \times \sin(2\pi H_0 t)$

2.2 New Terms: Osc_term and a_ExpFreq

Osc_term — Standing wave oscillation (FIRST in UQFF):

$$g_{\mathrm{Osc_term}}(r,t) = A_{\mathrm{osc}} \cos(k_{\mathrm{osc}} r) \sin(\omega_{\mathrm{osc}} t)$$

With A_{osc} = system-dependent amplitude, $k_{\mathrm{osc}} = 2\pi/r_{\mathrm{char}}$, $\omega_{\mathrm{osc}} = 2\pi f_{\mathrm{char}}$.

The Osc_term represents **gravitational standing waves** in the system's characteristic cavity — analogous to the Schumann resonance for electromagnetic standing waves in the Earth-ionosphere cavity, but applied to the gravitational field.

a_ExpFreq — Expansion-frequency coupling (FIRST in UQFF):

$$a_{\mathrm{ExpFreq}}(t) = H_0 \cdot c \cdot \sin(2\pi H_0 t)$$

$$= 2.27 \times 10^{-18} \times 3 \times 10^8 \times \sin(2\pi \times 2.27 \times 10^{-18} t)$$

$$= 6.81 \times 10^{-10} \times \sin(1.427 \times 10^{-17} t) \text{ m/s}^2$$

Period: $T_{\mathrm{ExpFreq}} = 1/H_0 = 4.41 \times 10^{17} \text{ s} = 13.97 \text{ Gyr}$ (Hubble time). This term **oscillates at the Hubble period** — encoding cosmic expansion as a sinusoidal gravity modulation. At present epoch ($t = t_H$), $a_{\mathrm{ExpFreq}} = 6.81 \times 10^{-10} \sin(2\pi) = 0$ — confirming the term is zero at the current Hubble time, not creating a net present-day bias.

3. Full Resonance Equation

$$g_{\mathrm{res}}^{(j)}(r,t) = g_{\mathrm{DPM}}^{(j)}(1 + H_z t)(1 - B/B_{\mathrm{crit}}) + \sum_{i=1}^{10} a_i^{(j)}(r,t)$$

4. getSolutions(t) Results for 7 Canonical Systems

At $t = 1 \text{ Gyr} = 3.156 \times 10^{16} \text{ s}$:

4.1 SGR 1745-2900 Magnetar

Term	Value (m/s ²)
g_DPM	3.716×10^{12}
a_THz	$\sim 7.26 \times 10^{24}$
a_AetherRes	$\sim 4.9 \times 10^6$
Osc_term	$\sim 1 \times 10^{-3}$ (oscillatory)
a_ExpFreq	$\sim -6.81 \times 10^{-10} \sin(14.27) \approx 4.1 \times 10^{-10}$
g_res total	$\sim 3.73 \times 10^6$ (after UQFF coupling factors)

4.2 Sagittarius A*

Term	Value (m/s ²)
g_DPM	$\sim 6.25 \times 10^1$
a_AetherFreq	$\sim 1 \times 10^{-2}$
a_FluidFreq	$\sim 10^{-15}$

a_ExpFreq	~4.1\$	\times\$	\$10^{-10}\$	
g_res total	~1.52			

4.3 Tapestry Starbirth

Term	Value (m/s ²)
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g_DPM	~2.65\$
P_outflow	~10 ⁻¹⁰
Osc_term	~10 ⁻¹³
g_res total	~1.02\$

4.4 Universe Guide

Term	Value (m/s ²)
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g_DPM	~5.88\$
g_DM	~1.58\$
a_ExpFreq	~4.1\$
g_res total	~1.14\$

5. Term Hierarchy Across 7 Systems

Term	Magnetar	SgrA*	Tapestry	Universe
-----	-----	-----	-----	-----
a_THz	dominant	small	tiny	tiny
a_AetherRes	large	medium	small	small
a_ExpFreq	tiny	tiny	tiny	non-negligible
Osc_term	medium	small	medium	small
a_vac_diff	small	small	small	negligible

Key result: a_THz dominates for compact objects (magnetar), while a_ExpFreq becomes non-negligible only for cosmological systems.

6. Standard Model Comparison

Feature	SM	UQFF PAPER_458
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Resonance terms in gravity	None	10-term acceleration suite
Hubble oscillation	Not in gravity	a_ExpFreq = H0c sin(2\$pi\$H0t)
Standing-wave gravity	Not in gravity	Osc_term = A cos(k r) sin(\$omega\$t)
Multi-system side-by-side	Separate codes	getSolutions(t) for all 7

7. Testable Predictions

1. **a_ExpFreq period = Hubble time:** At $t=t_H$, $a_{\text{ExpFreq}} = 0$. At $t = t_H/4$, a_{ExpFreq} is maximum. CMB power spectrum $P(k)$ should show subtle periodic modulation with period corresponding to $T_{\text{ExpFreq}} = 1/H_0$.
2. **Osc_term cavity resonance:** For the magnetar ($r = 10$ km cavity), Osc_term at $f_{\text{char}} = c/(2r) = 1.5 \times 10^{10}$ Hz. Detectable as sub-millisecond periodic gravity wave from neutron star surface modes.
3. **a_THz universality:** For all compact objects, $a_{\text{THz}} \propto c^3/(GMr) \times f_{\text{THz}}^2$ — implies $a_{\text{THz}}/g_{\text{DPM}} = (c/v_{\text{escape}})^2 \times (f_{\text{THz}} r/c)^2$, a universal ratio testable via GW observations.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected $M-\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \left(\frac{G M}{r_H^2}\right)\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left(1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right)$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

$M-\sigma$ correction (PAPER_1048): The phonon-corrected $M-\sigma$ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac, [SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2} m^2 \phi_B^2 + \frac{\lambda}{4!} \phi_B^4 + \kappa \cdot \rho_{\mathrm{vac, [SCm]}} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_B} = \nabla \cdot (\rho_{\mathrm{SCm}} \mathbf{v}) \cdot \mathbf{B} + \kappa B_{\mathrm{crit}} \partial_t \phi_B = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b, seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{\mathrm{Bi}_i}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_B} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac, [SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.119 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m³.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 23, \quad n_{\mathrm{channel}} = 17/26$$

Since $p_{\mathrm{DVP}} = 23$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **103 yr** (field decay quiescence):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.119	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 23$	PASS Sub-threshold
BSH layers	26 harmonic terms $j = 1\dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection	
κ decay	$\$5.0 \times 10^{-4}$ day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE g_{total} $\rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} M_{\text{env}}$	$L_X \geq 1037 \text{ erg/s}$	Chandra CXC	PASS Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	PASS UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day}$ $\rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

17 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,Bi\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}} q^{\{n^2\}} / (-q;q)_n^2$
- $\phi_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \text{Sum}_{\{n=1\}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \text{Sum } R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}} [1 + [SSq]\exp(-kappa_i*n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_Scm	0.99	SCm manifold completeness
rho_Scm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_Scm	$2*\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

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